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Marks Obtained: 9.5+4.5+4+6+6+9 = 39

Max. Marks: 40

Question 1 [CLO-2, C4]

You are building a machine learning system to classify different skin conditions based on clinical data. You decide to use binary classifiers to implement a multiclass classification system.

The dataset, consisting of four classes, with their respective sample counts and F1-scores is shown below. The total dataset size is 2320, and an 80:20 train-test split is used.

Class	Description	Number of Samples	Train F1 Score	Test F1 Score
0	Healthy	1000	0.97	0.80
1	Eczema	450	0.78	0.71
2	Psoriasis	200	0.66	0.60
3	Fungal Infection	670	0.89	0.87

1a How many binary classifiers will be required if you choose:

One-vs-Rest (OvR) approach?

Since, 4 classes, 4 classifiers will be required. (binary) 2

One-vs-One (OvO) approach?

here, $p = 4$
 we know that;
 $\frac{p(p-1)}{2} = \frac{4(4-1)}{2} = 6$ classifiers needed. (binary)

1b Compute weighted-average F1-score for the train set. Would the weighted-average F1-score be affected by class imbalance? Briefly justify your answer.

$$\text{Weighted Average F1 score} = \frac{(800 \times 0.97) + (360 \times 0.78) + (160 \times 0.66) + (536 \times 0.89)}{800 + 360 + 160 + 536}$$

$$\text{W A F1-score} = 0.88$$

class 0: 80% of 1000 = $0.8 \times 1000 = 800$

class 1: 80% of 450 = $0.8 \times 450 = 360$

class 2: 80% of 200 = $0.8 \times 200 = 160$

class 3: 80% of 670 = $0.8 \times 670 = 536$

Yes, F1-score (weighted average) will be affected as number of samples differ.
 Suppose, we have 100 samples of each for train.
 The F1-score will be 0.66 as F1 remain same.

1c Assuming the dataset to be tabular, suggest one way to increase the samples of the minority classes and one preprocessing step other than augmentation, along with its effect.

One preprocessing step other than augmentation is filtering removing duplicates. It will lessen the computation cost by removing redundancy. 2.5

To increase the samples of the minority classes, collect more data of the minority class. or synthetically create new samples for that class by changing feature values.

Question 2 [CLO-2, C4]

binary

A logistic regression classifier is trained to classify whether a tumor is malignant ($y = 1$) or benign ($y = 0$) based on two features: x_1 : Tumor Size (in cm), x_2 : Patient Age (in years)

The learned parameters (weights) for the logistic regression model are:

$$\theta_0 = 12 \quad \theta_1 = 0.6 \quad \theta_2 = 0.3$$

2a Find the equation of the decision boundary for the classifier, showing graphically the $y = 1$ and $y = 0$ regions.

$$(2) - z = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$$\text{Let } z = 0$$

$$0 = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

Putting values

$$0 = 12 + 0.6x_1 + 0.3x_2$$

$$-12 = 0.6x_1 + 0.3x_2$$

(1) This is decision boundary equation

$$x_1 = -\frac{(12 + 0.3x_2)}{0.6}$$

$$x_2 = 0$$

$$x_1 = -20$$

$$x_2 = 1, x_1 = -20.5$$

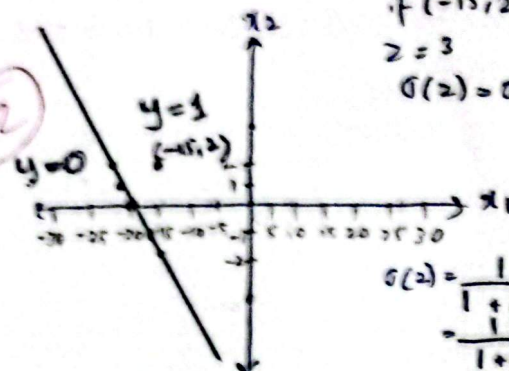
$$x_2 = -1, x_1 = -19.5$$

$$x_2 = 2, x_1 = -21$$

$$x_2 = -2, x_1 = -19$$

$$\text{Putting } x_2 = (-20, 0) \text{ in}$$

$$\text{in eq (2)} \quad z = 12 + 0.6(-20) + (0.3)(0) = 0$$



$$\text{if } (-15, 2): \\ z = 3 \\ \sigma(z) = 0.95 \approx 1$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} \\ = \frac{1}{1 + e^{-3}}$$

2b You evaluated the logistic regression model on a test set of 10 patients. The actual diagnoses and model-predicted classes (based on a threshold of 0.5) are shown below. Compute the precision of the classifier.

Patient ID	1	2	3	4	5	6	7	8	9	10
y	1	0*	1	0*	1*	0*	1	1	0*	0*
$\hat{y}$	1	0	1	1	0	0	1	1	0	1

$$TP = 4$$

$$TN = 3$$

$$FP = 2$$

$$FN = 1$$

$$\text{Precision} = \frac{TP}{TP + FP} \Rightarrow \frac{4}{4 + 2} \Rightarrow 0.67 = \text{Precision}$$

2c Suggest a way through which this precision can be improved without retraining the model. Briefly justify your answer.

By ~~adding~~ removing the FP sample $FP = 0$ and $\frac{TP}{TP} = 1$. This will improve the model.

Question 3 [CLO-1, C3]

Give two major differentiating point between:

3a Regularization and Normalization

- 1) regularization is the solution for overfitting whereas normalization is a technique of scaling
- 2) parameters are added in regularization whereas parameters are scaled in normalization.

3b Hyperbolic Tangent (Tanh) and Sigmoid activation functions

1) The formula for tanh of is $f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ and sigmoid of is $f(z) = \frac{1}{1 + e^{-z}}$

2) The derivatives of of tanh is $f'(z) = 1 - f(z)^2$ and for sigmoid is $f'(z) = f(z)(1 - f(z))$.

Value Range

3) tanh $f(z) = [-1, 1]$

Sigmoid $= [0, 1]$

Question 4 [CLO-1, C3]

Consider the following regularized logistic regression cost function:

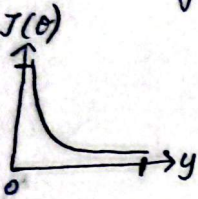
$$J(\theta) = -\frac{1}{n} \sum_{i=1}^n [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})] + \frac{\lambda}{2n} \sum_{j=1}^m \theta_j^2$$

- 4a Explain the three main terms in this cost function, as indicated by the horizontal lines.

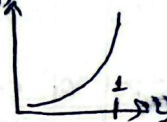
3

- 1) The cross entropy cost function when $y=1$
- 2) The cross entropy cost function when $y=0$
- 3) Ridge Regularization.

For explanation of part (1)
The cost func. reduced
when y reaching 0.1
This part represent that.



Explanation of part (2)
The cost function increases when reaching y towards 1
This part represents that the $J(\theta)$



Explanation of part (3):
This is the solution for overfitting.

- 4b Which variant of gradient descent is used in this formulation?

1

Batch gradient descent.

- 4c Which type of regularization is used here? Can it cause feature reduction?

2

Ridge regularization is used.
No, it does not reduce feature.

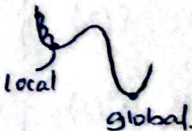
Question 5 [CLO-2, C4]

Comment if true or false, and justify briefly.

- 5a With optimal hyperparameter settings, gradient descent algorithm guarantees achievement of global minima.

2

Gradient descent algorithm may assume $m=0$ (slope) of local minima as gradient global minima. achieved.
Hence, this statement is not true. $\downarrow$ and may stop there.



- 5b Gradient descent algorithm guarantees optimal solution for Linear Regression.

2

No, gradient descent algo doesnot guarantee optimal solution for linear regression. For example, by large learning rate, there are chances, that the algo. miss the global minima.

- 5c In k-fold cross validation, the average test error of all folds is reported as the final test error to the client.

2

Yes.

We find the average validation error but hands the client the final test error.

3

Question 6 [CLO-1, C3]

Consider an artificial neural network with the following architecture:

- Input layer (layer 0): 2 neurons
- Hidden layer (layer 1): 2 neurons using the ReLU activation function
- Output layer (layer 2): 2 neuron using the Sigmoid activation function

6a Draw the network, labelling all the nodes properly. Add bias nodes where applicable.

Consider the following inputs to this network:

Input vector excluding bias: $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ Target output: $y = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

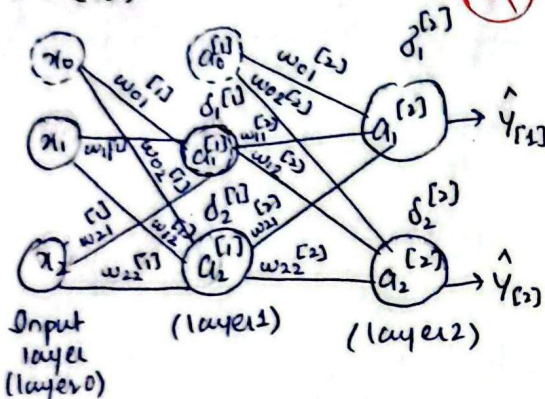
Initial weight matrices including biases: $w^{(1)} = \begin{bmatrix} 0.1 & -0.3 \\ 0.4 & 0.2 \\ 0.1 & -0.5 \end{bmatrix}$, $w^{(2)} = \begin{bmatrix} 0 & 0 \\ 0.3 & -0.3 \\ -0.1 & 0.1 \end{bmatrix}$

6b Compute the output of the hidden layer.

6c Compute the outputs of the network $\hat{y}_{[1]}$ and $\hat{y}_{[2]}$.

6d How many error values needs to be calculated for this network in each backward pass?

Part (6a):



Part (6b)

Output of hidden layer:
Neurons using ReLU.

$$f(z) = \begin{cases} 0 & z \leq 0 \\ z & z > 0 \end{cases}$$

$$f'(z) = \begin{cases} 0 & z \leq 0 \\ 1 & z > 0 \end{cases}$$

$$\sqrt{x_0} = 1$$

$$z_1^{(1)} = x_0 w_{01}^{(1)} + x_1 w_{11}^{(1)} + x_2 w_{21}^{(1)}$$

$$= 1(0.1) + (1)(0.4) + (0)(0.1)$$

$$z_1^{(1)} = 0.5$$

$$a_1^{(1)} = \text{since } z_1^{(1)} > 0$$

$$= \therefore z$$

$$a_1^{(1)} = 0.5$$

$$z_2^{(1)} = x_0 w_{02}^{(1)} + x_1 w_{12}^{(1)} + x_2 w_{22}^{(1)}$$

$$= (1)(-0.3) + (1)(0.2) + (0)$$

$$z_2^{(1)} = -0.1$$

$$a_2^{(1)} = z < 0$$

$$a_2^{(1)} = 0$$

$$a_0 = 1$$

Part 6(c):

Sigmoid being used.

$$f(z) = \frac{1}{1 + e^{-z}}$$

$$f'(z) = f(z)(1 - f(z))$$

Output of network $\hat{y}_{[1]}$

$$z_{[1]} = a_0 w_{01}^{(2)} + a_1 w_{11}^{(2)} + a_2 w_{21}^{(2)}$$

$$= (1)(0) + (0.5)(0.3) + (0)(-0.1)$$

$$z_{[1]} = 0.15$$

$$\hat{y}_{[1]} = \frac{1}{1 + e^{-0.15}} = 0.54$$

$$z_{[2]} = a_0 w_{02}^{(2)} + a_1 w_{12}^{(2)} + a_2 w_{22}^{(2)}$$

$$= (1)(0) + (0.5)(-0.3) + (0)(0.1)$$

$$z_{[2]} = -0.15$$

$$\hat{y}_{[2]} = \frac{1}{1 + e^{-(-0.15)}} = 0.46$$

Part (6d)

Four error values i.e. $\delta_1^{(2)}, \delta_2^{(2)}, \delta_1^{(1)}, \delta_2^{(1)}$ needs to be calculated.